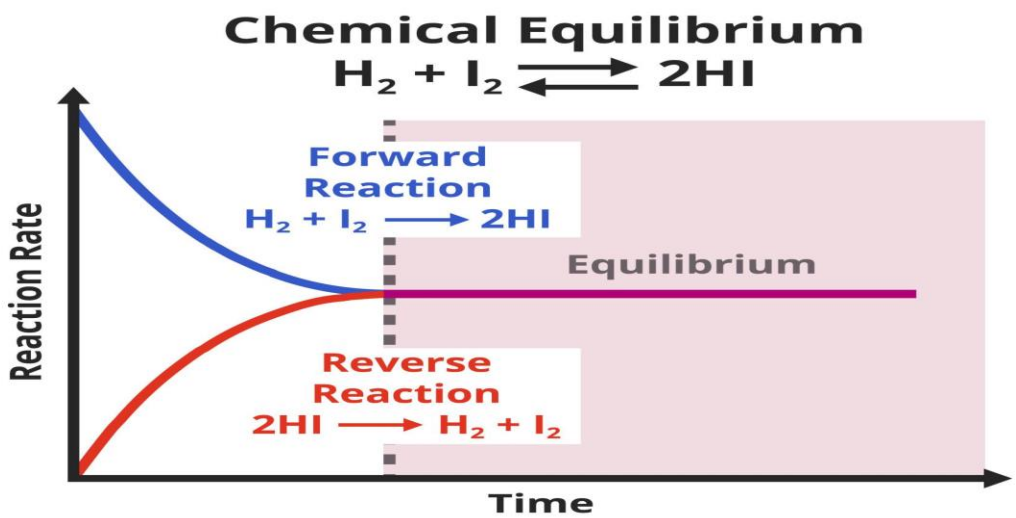
⍰ EQUILIBRIUM

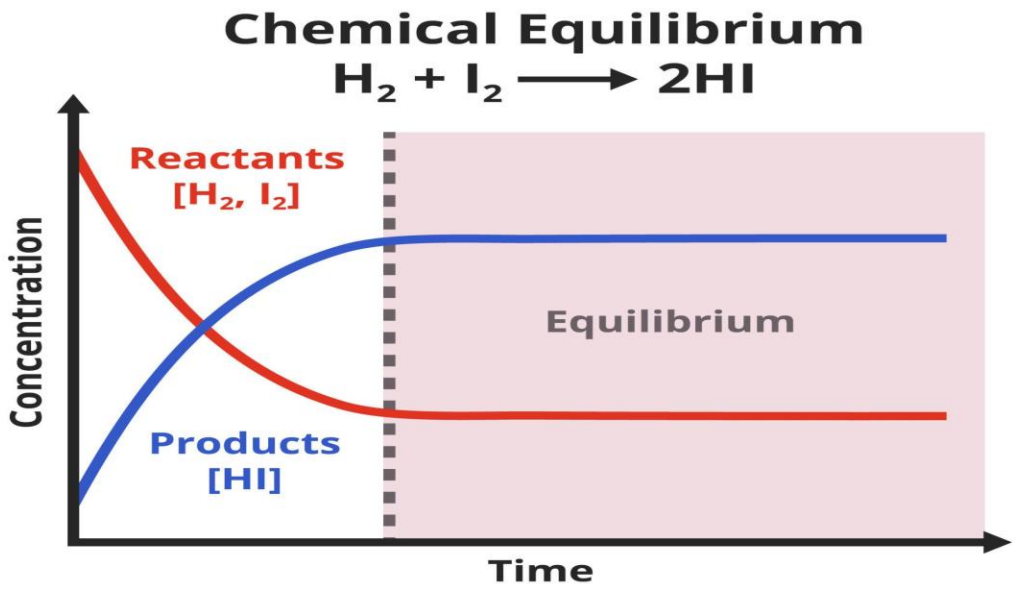
Introduction

Equilibrium is the state of a reversible reaction where the rate of forward reaction equals the rate of backward reaction. At equilibrium, the concentrations of reactants and products remain constant, although reactions continue at the molecular level. This chapter is important for NEET as it involves both conceptual understanding and numerical problems.

Dynamic Nature of Equilibrium

Chemical equilibrium is dynamic in nature because both forward and reverse reactions occur simultaneously. Even though there is no visible change, the reactions continue at equal rates.





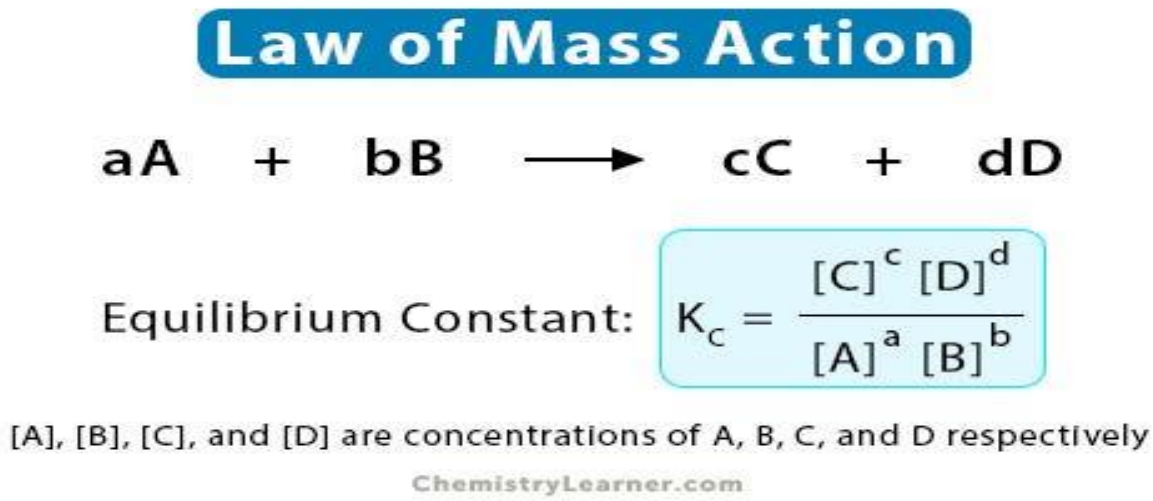
Law of Mass Action

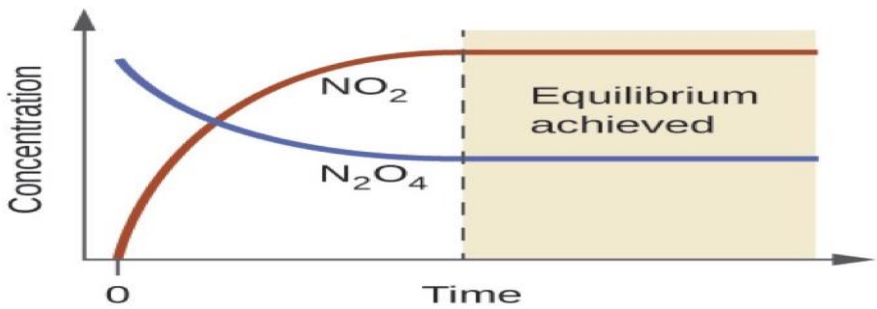
The law of mass action states that the rate of a chemical reaction is proportional to the product of the active masses of reactants. It forms the basis for the equilibrium constant.

Equilibrium Constant

The equilibrium constant is the ratio of concentrations of products to reactants at equilibrium, each raised to the power of their coefficients.

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$





It indicates the extent of a reaction.

Relationship between Kp and Kc

For gaseous reactions, equilibrium constants can be expressed in terms of pressure and concentration.

$$K_p = K_c \times (RT)^{\Delta n}$$

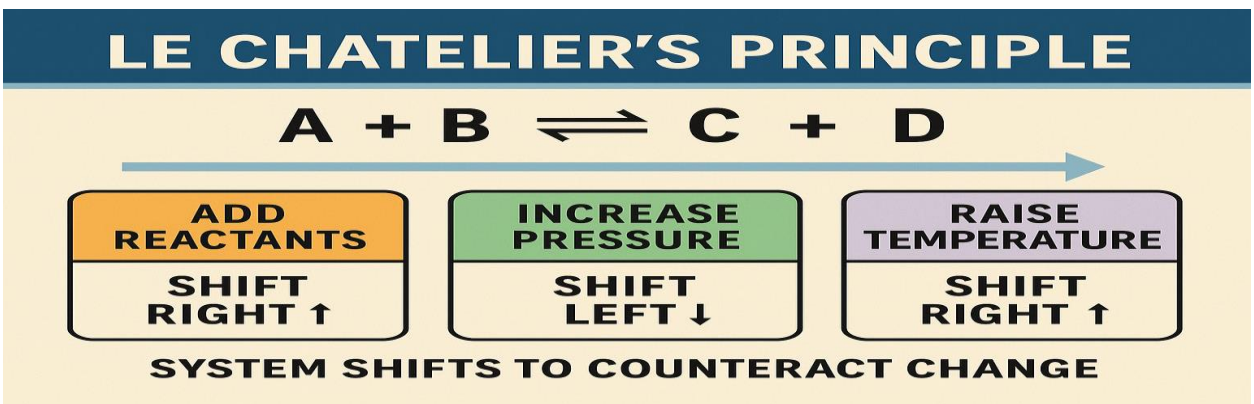
$$K_p = \frac{(P_C)^2}{(P_A) \times (P_B)}$$

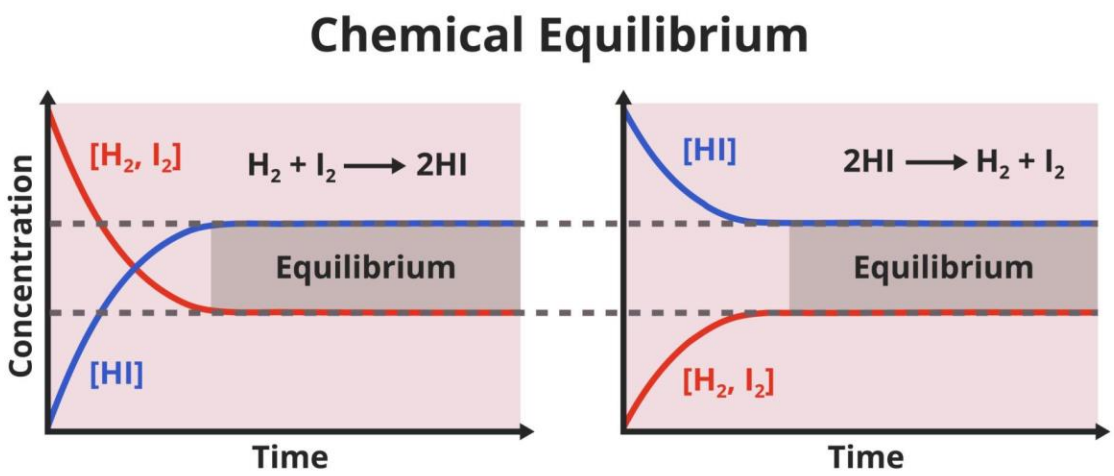
P_A = Partial Pressure of A
P_B = Partial Pressure of B
P_C = Partial Pressure of C

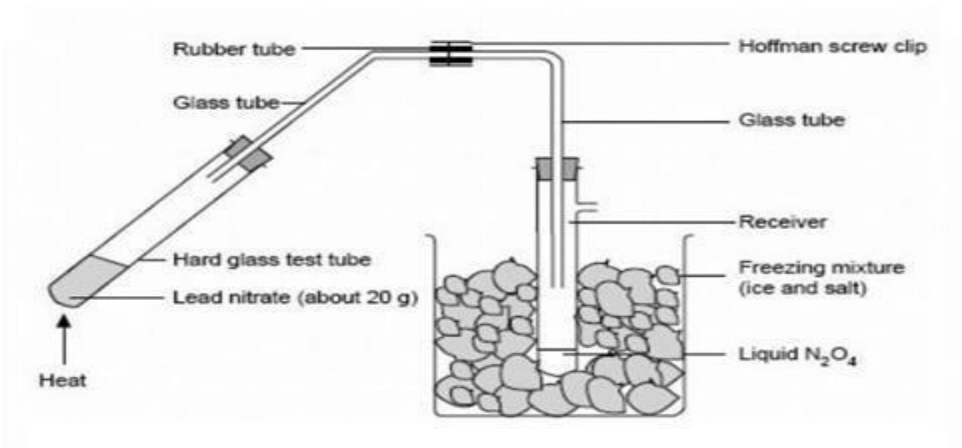
This relation is important for solving numerical problems.

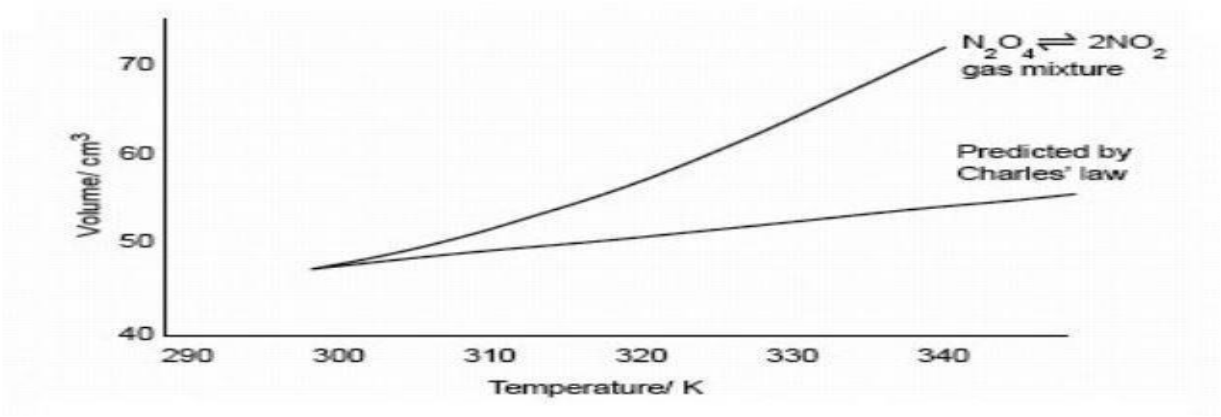
Le Chatelier's Principle

Le Chatelier's principle states that if a system at equilibrium is disturbed by changes in temperature, pressure, or concentration, it shifts to counteract the disturbance.









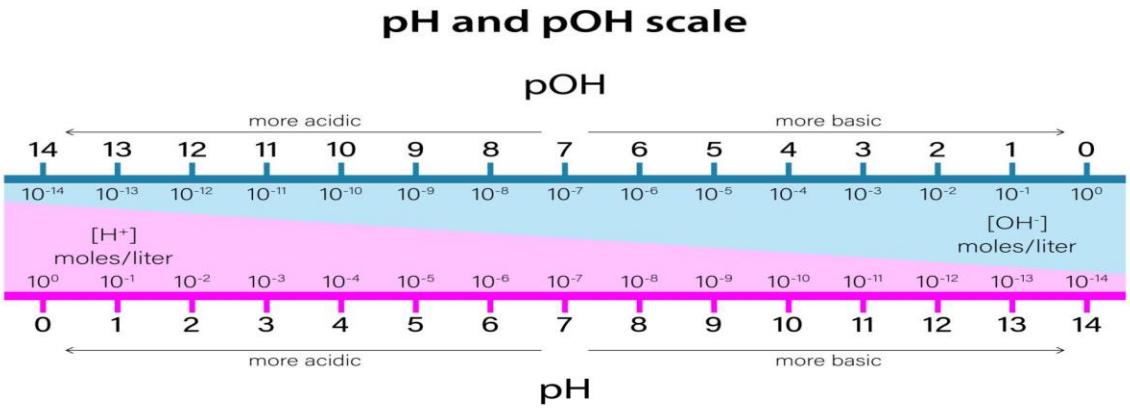
This principle helps predict the direction of equilibrium shift.

Acids and Bases

Acids are substances that donate protons, while bases accept protons. The strength of acids and bases depends on their ability to ionize in solution.

pH Scale

The pH scale measures the acidity or basicity of a solution. It is defined as the negative logarithm of hydrogen ion concentration.



***Q.44. Derive the relation pH + pOH = 14.**

Ans: Relationship between pH and pOH:

The ionic product of water is given as:

$K_w = [H_3O^+][OH^-]$

Now, $K_w = 1 \times 10^{-14}$ at 298 K

Thus, $[H_3O^+][OH^-] = 1.0 \times 10^{-14}$

Taking logarithm of both the sides, we write

$\log_{10}[H_3O^+] + \log_{10}[OH^-] = -14$

$-\log_{10}[H_3O^+] + \{-\log_{10}[OH^-]\} = 14$

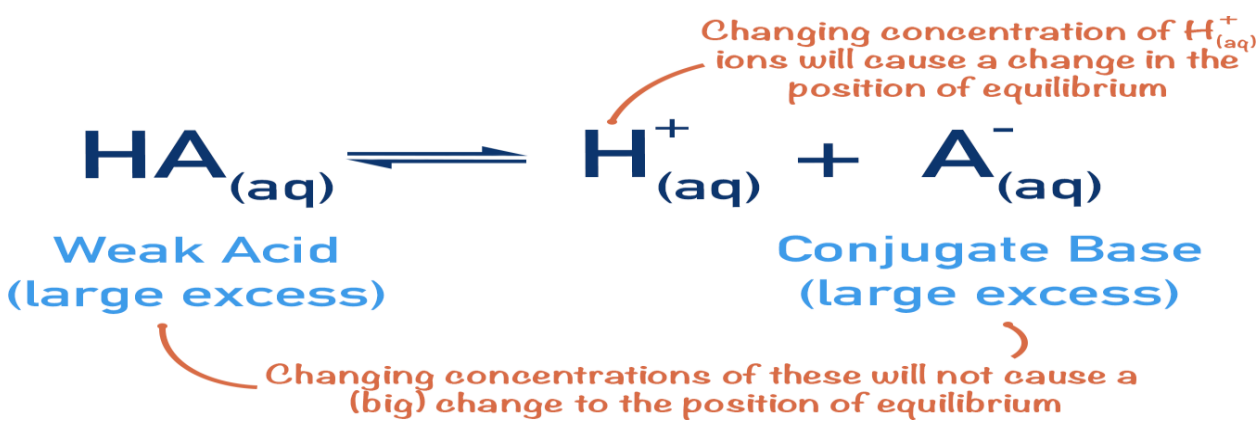
Now, $pH = -\log_{10}[H_3O^+]$ and $pOH = -\log_{10}[OH^-]$

$\therefore$ **pH + pOH = 14**

Lower pH indicates higher acidity, while higher pH indicates basicity.

Buffer Solutions

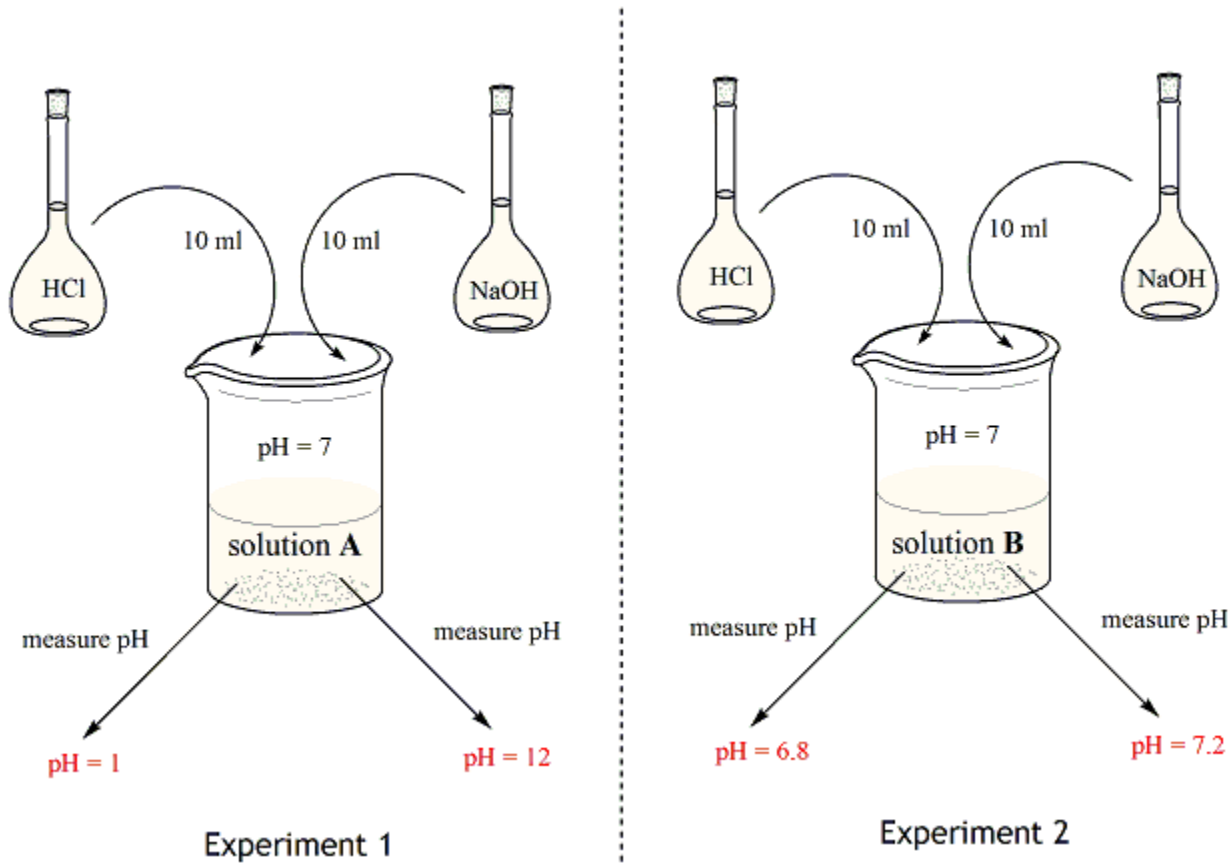
Buffer solutions resist changes in pH when small amounts of acid or base are added. They are important in biological systems and industrial processes.



$$HA\,(aq) + H_2O\,(l) \rightleftharpoons H_3O^+\,(aq) + A^-\,(aq)$$

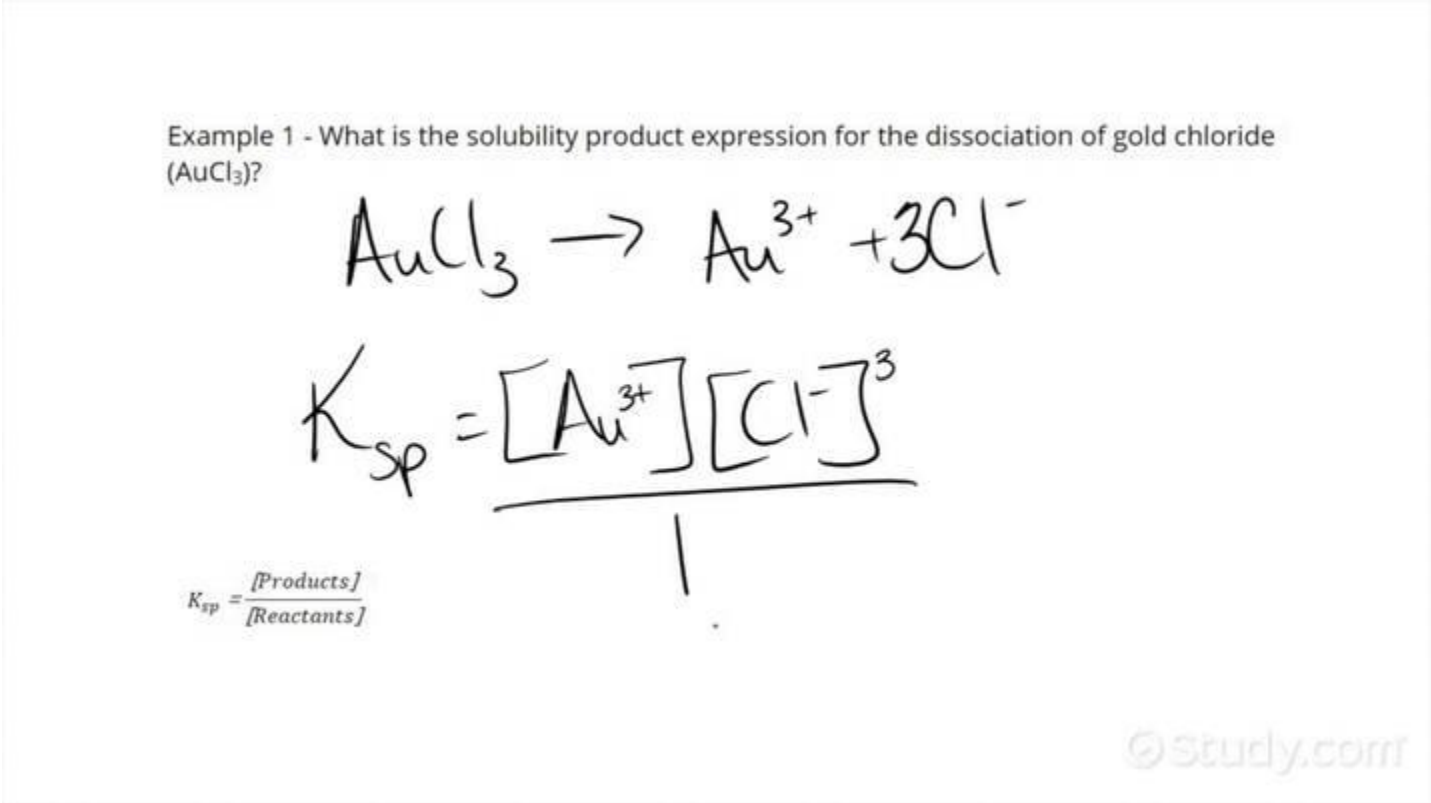
Henderson-Hasselbalch Equation

$$pH = pK_a + \log \frac{[base]}{[acid]}$$



Solubility Product

The solubility product constant (Ksp) represents the equilibrium between a solid and its ions in solution.



It is used to predict precipitation of salts.